

LLM-guided headline rewriting for clickability enhancement without clickbait

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Abstract

Enhancing reader engagement while preserving informational fidelity is a central challenge in controllable text generation for news media. Optimizing news headlines for reader engagement is often conflated with clickbait, resulting in exaggerated or misleading phrasing that undermines editorial trust. We frame clickbait not as a separate stylistic category, but as an extreme outcome of disproportionate amplification of otherwise legitimate engagement cues. Based on this view, we formulate headline rewriting as a controllable generation problem, where specific engagement-oriented linguistic attributes are selectively strengthened under explicit constraints on semantic faithfulness and proportional emphasis.

We present a guided headline rewriting framework built on a large language model (LLM) that uses the Future Discriminators for Generation (FUDGE) paradigm for inference-time control. The LLM is steered by two auxiliary guide models: (1) a clickbait scoring model that provides negative guidance to suppress excessive stylistic amplification, and (2) an engagement-attribute model that provides positive guidance aligned with target clickability objectives. Both guides are trained on neutral headlines drawn from a curated real-world news corpus. At the same time, clickbait variants are generated synthetically by rewriting these original headlines using an LLM under controlled activation of predefined engagement tactics.

By adjusting guidance weights at inference time, the system generates headlines along a continuum from neutral paraphrases to more engaging yet editorially acceptable formulations. The proposed framework provides a principled approach for studying the trade-off between attractiveness, semantic preservation, and clickbait avoidance, and supports responsible LLM-based headline optimization in journalistic settings. The guide models transfer to entirely human-authored clickbait corpora (AUROC 0.955 on Chakraborty 2016); a four-cell ablation on 300 held-out headlines shows the two guides exert the designed, independent, opposite effects; and against the DExperts and GeDi decoding-time baselines the method is Pareto-dominant, reaching comparable attribute control at markedly higher fidelity and fluency.

Keywords: Large Language Models; Controllable text generation; Clickbait mitigation; Inference-time control; FUDGE; Engagement optimization

1. Introduction

Digital news headlines function as the primary entry point to journalistic content and strongly influence readers' decisions about what to consume. In contemporary, algorithmically mediated information environments, headlines must attract attention within a narrow cognitive window while remaining faithful to the substance of the underlying article. This creates a structural tension between engagement and editorial integrity. In practice, this tension has encouraged the

use of clickbait-like strategies, which can increase short-term attention but are often associated with reader frustration, unmet expectations, and **long-term erosion of trust in news outlets [9]**.

Existing research has largely addressed this problem through clickbait detection, classification, and filtering, implicitly treating it as a discrete, undesirable stylistic category. Headlines are typically labeled as clickbait or non-clickbait, and models are trained to identify lexical, syntactic, or pragmatic markers **correlated with exaggeration or deception [1,2,3]**. While this work has produced useful datasets and benchmarks, it obscures an important distinction: many linguistic devices associated with clickbait, such as curiosity induction, emphasis framing, specificity manipulation, or interrogative constructions, are standard rhetorical tools in journalism. These mechanisms are not inherently misleading; they become problematic only when their use is amplified beyond editorially defensible bounds.

To make this distinction explicit, we define an engagement-attribute rubric that separates clickbait-oriented from clickability-oriented realizations of common headline mechanisms. The full comparison is presented later in Table 1 (Section 3.1.1), where these mechanisms are operationalized for synthetic data generation and controllable guidance. This comparison motivates a shift from binary classification to a continuous, controllable view of engagement. We therefore adopt a generative rather than purely discriminative perspective and posit two core assumptions. First, clickbait and clickability differ in degree rather than kind: clickbait arises when otherwise legitimate engagement mechanisms are amplified beyond proportionate or editorially acceptable levels. Second, engagement can be decomposed into controllable linguistic attributes, allowing individual mechanisms (e.g., curiosity, specificity, or framing) to be selectively strengthened or attenuated while maintaining explicit constraints on semantic fidelity and overall emphasis. From this perspective, the central challenge is not to eliminate engagement mechanisms, but to regulate their strength and interaction in a principled manner. This reframes headline optimization as a controlled text-generation problem, where engagement is modeled as a continuous, multidimensional space rather than a binary label. Such a formulation naturally motivates the use of large language models (LLMs) as flexible generators augmented with explicit control signals. **This rubric extends prior attribute-based clickbait attribution work [11] by explicitly pairing clickbait and clickability interpretations of each mechanism, and a dedicated human validation of the rubric is reported in Section 4.6. (the cross-reference now points to Section 4.6; mean Fleiss kappa 0.75 on a 150-headline sample, LLM-simulated proxy).**

In this work, we introduce a controllable headline rewriting framework that operationalizes this view **using Meta-Llama-3-8B-Instruct as the base generator**, guided at inference time by auxiliary models. The LLM is steered by two complementary signals: a BERT-based clickbait scoring model that provides negative guidance to penalize excessive stylistic amplification, and a BERT-based engagement attribute model that provides positive guidance aligned with specified clickability objectives. By integrating these signals within a guided decoding paradigm, the framework enables fine-grained control over both the degree and type of engagement expressed in rewritten headlines, supporting responsible engagement enhancement without semantic distortion or disproportionate emphasis.

2. Related Work

2.1 Clickbait Detection

Research on clickbait detection has progressed from feature-engineered classifiers to transformer-based and more interpretable frameworks. Early studies by Biyani et al. [1] and Chakraborty et al. [2] established that clickbait can be detected through lexical, stylistic, and structural cues, while Potthast et al.

[3] standardized the task through the Webis Clickbait Corpus 2017 and the Clickbait Challenge, enabling both binary and graded modeling of clickbait strength.

The field has since expanded in both scope and methodology. As reviewed in [4], clickbait research now spans not only detection algorithms, but also semantic manipulation, curiosity mechanisms, and credibility effects. Methodologically, recent work has shifted toward deep learning and pretrained language models. For example, [5] reports strong performance with a RoBERTa-Large-based detector and complements classification with LIME and SHAP for interpretability. At the same time, [6] extends the task to Urdu and highlights the value of sentence embeddings in low-resource settings.

Recent studies also explore more data-efficient and LLM-oriented paradigms. For instance, [7] shows that strong results can be achieved from headline text alone, even with very limited labeled data, whereas [8] finds that zero-shot and few-shot LLMs remain competitive and multilingual but still lag behind task-specific fine-tuned or prompt-tuned models. Together, these studies indicate that headline-level clickbait detection benefits from explicit supervision and task-adapted modeling rather than relying solely on general-purpose LLM inference.

At the same time, recent work suggests that clickbait should not be treated as a purely binary phenomenon. For example, [9] shows that engagement relationship with headline concreteness, suggesting that headlines can be too vague or too explicit. Similarly, [10] demonstrates that explicit informativeness features improve both performance and interpretability. Finally, [11] moves beyond binary detection toward a tactic-level explanation by separating clickbait detection from rhetorical-strategy attribution. Collectively, this line of work motivates treating clickbait as a graded, explainable, and attribute-driven phenomenon, which directly aligns with the present paper’s controllable rewriting perspective.

2.2 Controllable Text Generation

In parallel with advances in detection, a growing body of work has explored controllable text generation, aiming to steer language model outputs toward desired attributes such as style, sentiment, topic, or level of sensationalism. Early approaches emphasized training-time control. Conditional language models explicitly incorporate control variables during training, enabling the generator to condition its output on metadata or attribute labels. A prominent example is CTRL [12] (Keskar et al., 2019), a large-scale transformer trained with control codes that specify domains, styles, or other high-level properties. By learning these associations directly during pretraining, CTRL can generate text aligned with the provided control signal at inference.

Another training-time paradigm relies on reinforcement learning, where generation policies are optimized using reward functions that encode target attributes. While effective, such approaches are often computationally expensive and tightly coupled to specific control objectives.

More recently, research has increasingly focused on inference-time control methods that operate on a fixed, pretrained language model, avoiding full retraining. These approaches augment the decoding process with auxiliary models or signals that bias token selection. Plug-and-Play Language Models [13] exemplify this strategy by combining a frozen generator with lightweight attribute classifiers. During

generation, gradients from the classifier modify the language model's hidden states at each step, enabling control over attributes such as topic or sentiment without altering the base model's parameters.

GeDi [14] adopts a probabilistic perspective, using smaller generative models as discriminators. At each decoding step, GeDi computes class-conditioned probabilities for the desired attribute and its complement, applying Bayes' rule to reweight the token distribution of a larger generator. Similarly, DExperts [15] frames control as a product-of-experts problem, combining an "expert" model trained on desirable text with an "anti-expert" trained on undesirable text, so that only tokens favored by the expert and disfavored by the anti-expert are promoted.

FUDGE [16] provides a particularly elegant formulation for decoding. It trains a discriminator to predict whether a partially generated sequence will satisfy a target attribute. During decoding, the base language model's next-token probabilities are multiplied by the discriminator's estimate, yielding a principled Bayesian factorization of $P(\text{text}|\text{attribute})$ at inference time. This allows attribute control, such as style or formality, without modifying or fine-tuning the underlying generator.

Collectively, these methods demonstrate that fine-grained controllability can be achieved by augmenting pretrained language models at inference. In particular, approaches such as FUDGE highlight how lightweight, modular discriminators can steer fluent generation toward desired attributes through probabilistically grounded mechanisms, making them well-suited for applications that require subtle control, such as generating curiosity-inducing yet non-clickbait headlines, while preserving linguistic quality.

3. Method

3.1 Construction of Guided Control Models

To enable fine-grained control over engagement during headline rewriting, we construct two auxiliary guide models that operate alongside the base LLM at inference time: a clickbait-scoring model and an engagement-attribute model. Both models are trained on a synthetically generated dataset designed to encode proportional versus excessive realizations of engagement mechanisms explicitly. The construction process consists of four stages.

3.1.1 Engagement Attribute Rubric

We first define a structured rubric of engagement attributes commonly found in news headlines, including information-gap control, emphasis intensity, emotional framing, salience allocation, referential clarity, structural emphasis, and interrogative framing (Table 1). For each attribute, the rubric specifies paired interpretations corresponding to clickability-oriented and clickbait-oriented usage. These paired interpretations should be understood not as different surface forms, but as different uses of the same underlying rhetorical mechanism. For example, information-gap control may appear either as calibrated partial disclosure that invites interest while preserving the core claim, or as withholding of essential facts that manufactures unresolved curiosity. Similarly, emphasis intensity may range from proportional emphasis to overstatement of its importance. Across the remaining attributes, the same principle applies: clickability reflects bounded, content-faithful use of engagement devices, whereas clickbait reflects disproportionate or misleading use. This rubric provides an explicit operationalization of engagement as a set of controllable linguistic dimensions rather than an undifferentiated stylistic label. Accordingly, the rubric is used both to define the attribute configurations employed in synthetic headline generation and to guide supervision of the auxiliary engagement model. It serves as the foundation for both data generation and model supervision.

Attribute	Clickbait Interpretation	Clickability Interpretation
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Information gap control	Curiosity gap Deliberate withholding of essential facts to compel a click by creating unresolved curiosity	Information gap calibration Partial disclosure that invites interest while preserving the core factual claim
Emphasis intensity	Exaggeration Overstating importance, impact, or certainty beyond what is supported by the article	Proportional emphasis scaling Adjusting linguistic intensity to match the true significance of the event
Emotional framing	Emotional trigger Manipulative activation of fear, outrage, or excitement disproportionate to the facts	Affective salience framing Highlighting emotionally relevant aspects implied by the content without inflation
Salience allocation	Sensationalism Focusing on dramatic or marginal aspects at the expense of informational balance	Salience-oriented framing Foregrounding the most newsworthy element while maintaining contextual integrity
Structural emphasis	Lists or superlatives Use of absolute rankings or hyperbolic enumeration to imply exceptionalism	Structured highlighting Bounded use of ordering or comparison to organize key information
Referential clarity	Ambiguous references Vague entities or pronouns that obscure meaning to provoke curiosity	Referential under-specification Temporary abstraction that is resolved through reading without misleading
Reader addressing	Direct appeals Imperative or coercive language explicitly urging the reader to click	Reader relevance cues Implicit signaling of why the story matters to the reader
Narrative structure	Unfinished narratives Artificial cliffhangers that omit essential outcomes	Narrative continuation cues Indicating that further context follows while retaining essential information
Conceptual framing	Unexpected associations Forced or misleading linkage between unrelated concepts	Cross-domain framing Novel but factually grounded connections that increase interpretive interest
Interrogative form	Provocative questions Questions implying false premises or exaggerated stakes	Interrogative framing Questions used to frame inquiry or uncertainty without presupposition

Table 1: Clickbait vs. Clickability Interpretations of Common Headline Attributes

3.1.2 Synthetic Dataset Generation with Controlled Clickbait Interpretation

Using the engagement rubric, we generate a synthetic corpus of clickbait headlines via attributed prompting of a large language model (GPT-4o-mini), starting from a curated collection of real, neutral news headlines drawn from the True.csv portion of the ISOT Fake News Dataset [17,18]. Because this subset

was collected from Reuters.com, we treat these headlines as neutral, fact-oriented baseline formulations and do not generate additional neutral variants. Each source headline defines one data sample and is assigned a non-clickbait label with an all-zero engagement attribute vector.

For every source headline, we generate a single clickbait variant by prompting the model to rewrite it using a controlled subset of engagement attributes. Specifically, the prompt allows the activation of up to three clickbait-oriented engagement mechanisms (e.g., Curiosity, Exaggeration, Emotional Triggers), which are explicitly specified at generation time. The generation is constrained to exaggerate only the selected dimensions while preserving the original headline's underlying factual content and core semantic meaning.

As a result, each generated sample consists of a headline text, a binary clickbait label, and a sparse binary engagement-attribute vector indicating which attributes were exaggerated. This construction yields a dataset in which neutral headlines and clickbait headlines represent two controlled extremes of the engagement space. The resulting data provides clean supervision for training both the binary clickbait scoring model and the engagement attribute model, while maintaining semantic alignment with the original curated headlines. The full generation prompt (including per-tactic definitions and constraints), batch size (15 headlines per API call), the up-to-three-tactic sampling rule, and the JSON self-repair retry policy (up to 3 attempts) are documented in Section 3.3. All prompt templates and generation parameters are also released with the code repository to support exact reproduction.

Table 2 shows representative examples from the synthetic dataset construction process. Each example pairs a real, curated neutral headline with an explicitly specified clickbait tactic and the resulting generated headline. The table illustrates how a fixed prompt template, differing only in the activated tactic, yields controlled, interpretable stylistic amplification. These examples demonstrate that the dataset encodes clickbait behavior at the level of individual engagement mechanisms, supporting both binary clickbait supervision and attribute-level modeling.

Neutral Source Headline	Attribute-Control Prompt (excerpt)	Generated Headline
<i>The White House budget chief expects a delay in hitting the debt limit.</i>	Rewrite the headline using the Curiosity Gap tactic. Deliberately withhold essential details to create unresolved curiosity. Do not add new facts.	<i>The White House budget chief reveals a shocking delay in the debt limit.</i>
<i>Lawsuit says North Carolina bathroom law still harmful.</i>	Rewrite the headline using Exaggeration . Overstate the importance and impact of the decision beyond what is directly supported by the article.	<i>You won't believe the impact of North Carolina's bathroom law.</i>
<i>Switzerland urges voters to keep the fee for public broadcasters.</i>	Rewrite the headline using an Emotional Trigger . Activate concern or alarm disproportionate to the reported findings.	<i>What remains hidden about Switzerland's public broadcaster fee debate</i>
<i>No country for migrant stowaways caught on ferry between Ukraine and Turkey.</i>	Rewrite the headline using Lists or Superlatives . Imply exceptionalism through hyperbolic ranking or enumeration.	<i>The dramatic truth behind migrant stowaways caught on a ferry</i>

Table 2: Examples of Attribute-Controlled Clickbait Generation via Explicit Tactics

Table 3 summarizes the composition of the synthetic dataset by engagement intensity. Neutral headlines, drawn directly from the curated source corpus, contain no activated engagement tactics, while clickbait variants are generated by activating one, two, or three tactics per headline. The dataset is approximately balanced across these groups, enabling the scoring model to learn a graded notion of

clickbait strength while preserving a clear distinction between neutral and increasingly amplified formulations.

Class	Number of Active Tactics	Number of Samples
Neutral	0	21667
Clickbait	1	7115
Clickbait	2	7255
Clickbait	3	7297

Table 3: Dataset composition by engagement intensity

3.1.3 Training the Clickbait Scoring Model

The clickbait scoring model is designed to operate on partial generations rather than on complete headlines alone, as it is used to guide the base language model during inference-time decoding. In this setting, the model must evaluate stylistic amplification from incomplete text, since guidance is applied incrementally as tokens are generated. Training a scorer solely on finished headlines would therefore be insufficient, as it would provide no supervision for intermediate prefixes.

To address this requirement, each headline in the synthetic dataset is decomposed into all of its token-level prefixes. Given a headline represented as a token sequence, $w_1, \dots, w_T$, we construct training inputs $w_1, \dots, w_t$ for all $t \in \{1, \dots, T\}$. Each prefix inherits the clickbait label of the full headline, allowing the model to learn how clickbait-related signals accumulate progressively as the headline unfolds. To explicitly distinguish complete headlines from prefixes, a terminal period token is appended to the full headline during training, while prefixes are left unmarked. This simple convention enables the model to condition its predictions on whether the input represents a completed headline or an intermediate generation state.

The scoring model itself is implemented by fine-tuning a BERT encoder with a binary classification head. For a given partial headline, the model outputs a scalar clickbait score, interpreted as the predicted probability that the current prefix belongs to the clickbait class. During guided decoding, this score serves as a negative control signal to penalize candidate continuations that increase the likelihood of clickbait. Because very short prefixes provide limited semantic evidence and are unreliable indicators of stylistic intent, we employ a length-aware weighted loss function during training. Loss contributions from shorter prefixes are down-weighted relative to those from longer prefixes and complete headlines, encouraging the model to focus on informative contexts while remaining robust to partial inputs. This training strategy yields a clickbait scorer that produces stable, prefix-consistent signals suitable for guiding controlled headline rewriting at inference time.

For training and evaluation, the data are split at the source headline level prior to synthetic generation to prevent leakage across prefixes. The curated set of real neutral headlines is first divided into 80% for training/validation and 20% for testing. The split is performed uniformly at random at the source-headline level and is not stratified by tactic combination; because every source headline contributes both a neutral sample and its clickbait variant, the resulting class balance is preserved by construction. Synthetic clickbait variants and their corresponding prefixes

are then generated separately within each split. Model fine-tuning is conducted for 2–3 epochs, with early stopping based on validation loss to prevent overfitting. Optimization uses the AdamW optimizer with a learning rate of 2×10^{-5} and linear decay thereafter. Mini-batches are constructed by sampling prefixes of varying lengths to ensure exposure to both partial and near-complete contexts within each batch.

On the held-out synthetic test set, the scorer achieved near-perfect discrimination between neutral and clickbait headlines and maintained consistently high accuracy across different numbers of activated tactics, which is expected given the controlled synthetic setup and the explicit rubric used to generate clickbait variants.

3.1.4 Training the Engagement Attribute Model

In parallel with the clickbait scoring model, we train an engagement attribute model to predict the presence, and when applicable, the strength, of specific engagement mechanisms defined in the rubric. The model is trained on the same synthetic dataset and leverages structured supervision from explicit attribute annotations. Whereas the clickbait scorer provides a global assessment of disproportionate amplification, the engagement model captures fine-grained, attribute-specific signals corresponding to distinct rhetorical tactics.

The engagement model is implemented by fine-tuning a pretrained BERT encoder with a multi-class, multi-label prediction head. Each attribute is modeled as an independent prediction dimension, allowing the model to estimate whether a given engagement mechanism is present in the input. This design supports simultaneous prediction of multiple attributes and enables selective activation during inference-time guidance. As the model is used to guide generation incrementally, training is performed on both complete headlines and all token-level prefixes derived from them.

To support prefix-level operation, each headline is decomposed into its prefixes, and all prefixes inherit the attribute annotations of the full headline. Complete headlines are marked with a terminal punctuation token, while prefixes remain unmarked, enabling the model to distinguish between finished and partial inputs. Training employs a length-aware weighted loss function, in which each example's contribution is scaled by its prefix length. Very short prefixes, which provide weak evidence for attribute realization, are assigned lower weights, while longer prefixes and complete headlines are emphasized.

(60%) (20%) for up to 3 epochs with early stopping based on validation loss, and for 2-3 epochs using the AdamW optimizer with a learning rate of 2×10^{-5} . In the full multi-label setting, the model achieved ... micro-F1 and ... macro-F1. For interpretability, the per-tactic analysis in Figure 1 focuses on single-tactic examples, which isolate each engagement mechanism, allowing examination of tactic-specific recognition quality and structured confusions without ambiguity due to overlapping labels. The dataset is split at the source headline level into training, validation, and test sets, with 80% allocated to training and validation and 20% to the test set to prevent leakage across prefixes. The split is uniform at random at the source-headline level and is not stratified by tactic vector. Fine-tuning is performed. During guided decoding, the engagement attribute model supplies positive, attribute-specific guidance signals that encourage the base LLM to realize selected engagement mechanisms. In contrast, the clickbait scoring model provides a complementary negative signal that constrains excessive amplification.

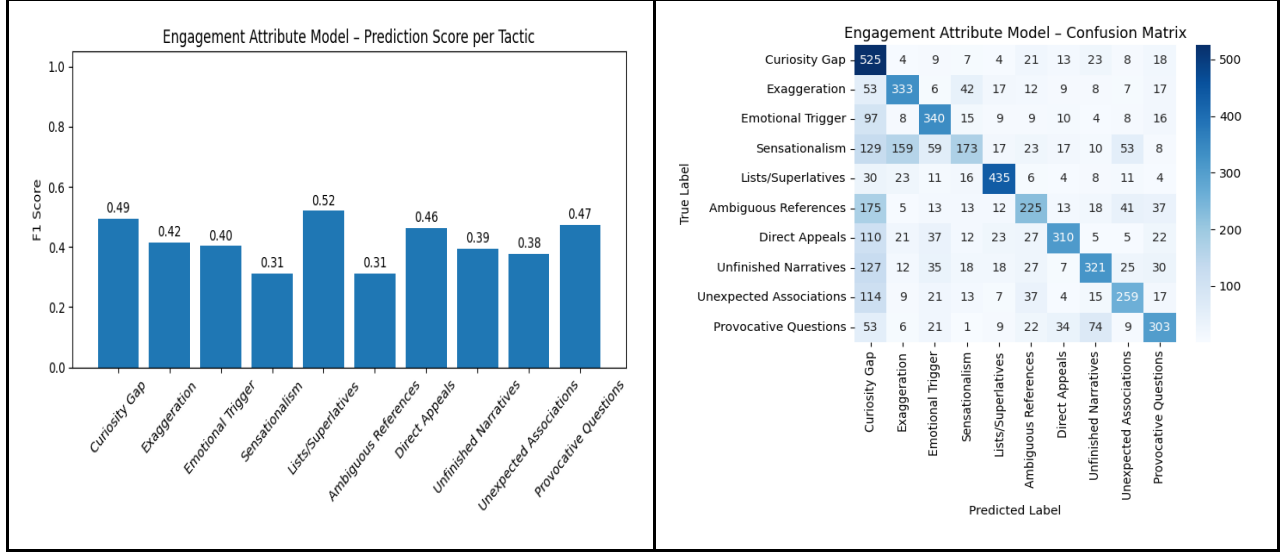


Figure 1: (Left) Per-tactic F1 score of the engagement attribute model on single-tactic test examples. (Right) Confusion matrix of predicted vs. targeted tactic on the same subset.

Figure 1 summarizes the behavior of the engagement attribute model through two complementary visualizations. Although the model is trained in a multi-label setting, we report per-tactic results on single-tactic examples for interpretability. This isolates each engagement mechanism, enabling analysis of tactic-specific recognition quality and structured confusions without the ambiguity introduced by overlapping labels. The bar plot shows differences in how reliably various engagement tactics are detected, with some attributes, such as Lists/Superlatives and Curiosity Gap, appearing more consistently recognizable. In contrast, others, including Sensationalism and Ambiguous References, are more difficult to identify.

The confusion matrix provides a more detailed view of the model’s predictions for single-attribute examples. A clear diagonal pattern indicates that the model often identifies the correct dominant tactic, while structured off-diagonal errors reveal confusion between semantically related attributes, such as emotional triggers and sensational phrasing. The results suggest that the model captures distinct engagement patterns while reflecting the natural overlap among persuasive strategies, providing informative and interpretable signals to guide controlled headline generation.

3.2 FUDGE-Based Controlled Generation with Positive and Negative Guidance

To enable fine-grained control over headline rewriting, we adopt a FUDGE-based (Future Discriminators for Generation) inference-time control framework, in which generation from a base large language model is steered by auxiliary scoring models operating on partial outputs. In this setting, guidance is applied incrementally during decoding by modifying token-selection probabilities based on predicted properties of the generated prefix.

Let $x_{1:t}$ denote the current generation prefix and y a candidate next token. At each decoding step, the base LLM proposes candidate continuations, which are then re-weighted using guidance scores derived from the auxiliary models evaluated on the extended prefix:

$$x_{1:t} \oplus y, \quad (1)$$

where $\oplus$ denotes the concatenation of the candidate token to the current prefix. The resulting guidance scores are then used to adjust the relative probabilities of candidate next tokens before sampling or selection.

3.2.1 Negative Guidance via Clickbait Scoring

The clickbait scoring model provides a negative control signal intended to suppress disproportionate stylistic amplification. For each candidate's continuation, the scorer outputs a scalar score:

$$S_{cb}(x_{1:t} \oplus y), \quad (2)$$

$S_{cb}(\cdot)$ representing the likelihood of clickbait. This score is incorporated as a penalty term with an associated weight λ_{neg} , reducing the probability of tokens that increase the predicted clickbait score. Intuitively, this signal acts as a soft constraint that discourages the generator from entering clickbait-prone regions of the engagement space without imposing hard rejection.

3.2.2 Positive Guidance via Engagement Attribute Signals

In parallel, the engagement attribute model provides positive control signals aligned with target clickability objectives. For each attribute k , the model produces a normalized score.

$$S_k(x_{1:t} \oplus y), \quad (3)$$

indicating the strength of that engagement mechanism. Desired attributes are assigned positive weights, while discouraged attributes are assigned negative weights. The overall positive guidance signal is computed as a weighted sum

$$S_{pos}(x_{1:t} \oplus y) = \sum_k \omega_k S_k(x_{1:t} \oplus y), \quad (4)$$

where $w_k \in \{-1, 0, 1\}$ encodes whether an attribute is encouraged, ignored, or actively suppressed. This formulation allows explicit specification of which engagement mechanisms should be strengthened and which should be avoided.

3.2.3 Combined Guidance Objective

The final token selection score combines the base LLM log-probability with both guidance terms: We use the canonical FUDGE factorization in the log domain: $\text{score}(y) = \log p_{LLM}(y) + \alpha * \log \text{tac}(x, y) + \beta * \log(1 - \text{cb}(x, y))$. The log domain replaces the additive-probability heuristic; on-weight magnitudes are re-tuned per objective (Section 4.4).

$$\log p(y|x_{1:t}) \propto \log p_{LLM}(y|x_{1:t}) + \lambda_{pos} S_{pos}(x_{1:t} \oplus y) - \lambda_{neg} S_{cb}(x_{1:t} \oplus y), \quad (5)$$

By adjusting the guidance weights $\lambda_{neg} \geq 0$ and $\lambda_{pos} \geq 0$, the system can smoothly trade off engagement enhancement against editorial restraint.

This FUDGE-based formulation enables the base LLM to navigate the engagement space in a controlled, interpretable manner. Positive signals encourage the realization of selected clickability attributes, while negative signals continuously penalize excessive amplification. Together, these

complementary controls enable responsible headline rewriting that enhances reader appeal while preserving semantic fidelity and avoiding clickbait extremes.

3.3 Implementation Details and Reproducibility

Base generator: Meta-Llama-3-8B-Instruct loaded from HuggingFace in FP16. All reported experiments run on cloud GPUs (NVIDIA A10G / L40S) orchestrated with Modal. Decoding uses greedy rescoring over the top-k next-token candidates with $k = 50$. Under the log-domain objective of Section 3.2.3 the guidance weights are swept over α , β in $\{0, 1, 2, 4\}$ on a disjoint 40-item tuning subset, and the reported operating point is $\alpha = 4$ (engagement) and $\beta = 2$ (clickbait suppression), selected on independent metrics only. A linear warm-up multiplies α by $\min(\text{step}/6, 1.0)$ over the first six tokens, and an EOS-length bonus discourages headlines shorter than six or longer than thirty-two whitespace tokens. Maximum generation length is 150 tokens.

Guide models: two BERT-base encoders, each fine-tuned for 2–3 epochs with early stopping on validation loss, AdamW at learning rate 2×10^{-5} with linear decay, mixed prefix-length mini-batches, and a length-aware weighted loss that down-weights very short prefixes (<4 tokens) relative to full headlines.

Synthetic data generation: source headlines are drawn from the True.csv split of the ISOT Fake News Dataset (12,600 Reuters headlines), yielding 12,599 validated clickbait variants after JSON-validity and duplicate filtering, produced by GPT-4o-mini in batches of 15 at temperature 0.7 with 1-3 tactics per call. On the held-out synthetic split the clickbait guide reaches AUROC 0.9998 and the ten-label engagement guide macro-F1 0.52; their transfer to human-authored data is characterized in Sections 4.1-4.2. A separate DistilBERT detector trained only on the human-authored Chakraborty corpus is the independent clickbait evaluator used throughout Section 4.

Complete prompt templates for both the synthetic-clickbait generation and the FUDGE rewrite instruction, together with all training and decoding hyperparameters, are released with the public code repository referenced in the Data Availability section. Approximate wall-clock cost per rewritten headline is about 1.2 s on the A10G/L40S cloud configuration; every generation and evaluation artifact is stored for reproducibility.

4. Results

Table 4 presents qualitative examples of headline rewrites targeting different engagement attributes. The neutral headlines report events factually without rhetorical framing. When rewritten with engagement guidance but no clickbait brake, the generated headlines increase engagement through stronger rhetorical strategies (curiosity gaps, exaggeration, metaphor, emotional triggers, provocative questions), often with more dramatic language. When the clickbait brake is applied, the headlines remain engaging and continue to reflect the intended tactic, but the phrasing becomes more restrained and informative. These examples illustrate that the framework can express specific engagement attributes while moderating excessive sensationalism through the clickbait control mechanism. Quantitative results follow in Sections 4.1-4.6.

Target engagement attribute	Neutral headline	Rewriting (engagement guidance, no clickbait brake)	Rewriting (with clickbait brake)
Curiosity gap	<i>European football: Harry Kane breaks personal record, Dortmund suffer loss</i>	<i>What is behind Harry Kane's remarkable milestone in European football, as Dortmund's defeat adds to the drama?</i>	<i>European Football: Can Harry Kane's Unstoppable Form Breed a New Era of Success?</i>
Exaggeration	<i>Venezuelan crisis spawns boom in gambling.</i>	<i>Venezuela's Desperate Situation Ignites a Frenzy of High-Stakes Bets.</i>	<i>Venezuela's Desperate Struggle Highlights Troubled Citizens' Frenzied Quest for Luck.</i>
Unexpected associations	<i>Arctic drilling hits speed bumps in U.S. tax bill</i>	<i>Frozen ambitions stall in the unexpectedly icy waters of fiscal legislation.</i>	<i>Frozen ambitions stall in Washington's fiscal freeze.</i>
Curiosity Gap + Emotional Trigger	<i>Apple case exposes ongoing government rift over encryption policy</i>	<i>What is Behind the Apple Case: A Deepening Divide on Encryption Policy.</i>	<i>Behind Closed Doors: The Apple Case Reveals a Deepening Divide Over Encryption Secrets.</i>
Curiosity Gap + Provocative Questions	<i>Lockheed Martin wins \$450 million Pentagon contract, statement says.</i>	<i>What is Behind the Pentagon's \$450 Million Investment in Lockheed Martin?</i>	<i>Lockheed Martin Lands Lucrative Pentagon Deal: But at What Cost to Taxpayers?</i>

Table 4: Examples of headline rewrites targeting specific engagement attributes

4.1 Guide-model validation on external, human-authored corpora

To address the closed-loop concern that a scorer trained on GPT-4o-mini rewrites may detect generator artifacts rather than clickbait per se, we evaluate both guide models on two entirely human-authored corpora never seen in training: Chakraborty et al. 2016 (32,000 balanced headlines) and the Webis Clickbait Corpus 2017 (2,459 graded headlines). We also train a fully independent DistilBERT clickbait detector on the human-authored Chakraborty data alone, and use that independent detector, never our own guide, for every clickbait number reported below. All figures carry 95% percentile-bootstrap confidence intervals ($n = 1000$).

Table 5: Prefix clickbait guide on external human-authored corpora (95% bootstrap CI).

Corpus	n	AUROC	F1	graded Spearman
Chakraborty 2016 (binary)	32,000	0.955 (0.953-0.957)	0.858	
Webis-17 (graded)	2,459	0.740 (0.720-0.760)	0.512	0.426 (p~1e-109)

The guide generalizes strongly to clean human-authored clickbait (Chakraborty AUROC 0.955) and moderately but significantly to the noisier graded Webis corpus (AUROC 0.740; graded

Spearman 0.43). The engagement-attribute guide shows the same external validity: on Webis all ten tactics score significantly higher on human clickbait than non-clickbait (Spearman $r = 0.34-0.45$, every $p < 1e-60$). The guides transfer to data they never saw.

4.2 Prefix-level behavior on real headlines

Because the guide operates on partial prefixes during decoding, we validate it in that regime on real data: each labeled Chakraborty / Webis headline is truncated to the training word-ratios (0.3 / 0.5 / 0.7 / 1.0) and the prefix score is compared to the parent headline's human label. AUROC rises monotonically as the prefix grows (Chakraborty 0.891 at 30% of the headline to 0.954 at full length; Webis 0.695 to 0.739), and the separation widens monotonically (Chakraborty +0.62 to +0.75). The guide supplies a valid, monotone steering signal on real text from an early prefix.

4.3 Neutrality of the Reuters source pool

The independent human-authored detector was run over the full ISOT True.csv source pool (20,825 unique Reuters headlines). Only 0.61% exceed the 0.5 clickbait threshold (mean probability 0.007, median 0.0003), and the highest-scoring outliers are genuine listicle / question forms. This supports, rather than merely assumes, treating the Reuters pool as a neutral baseline.

4.4 Held-out rewrite evaluation and dual-guidance ablation

We evaluate rewrites on 300 held-out neutral headlines under the four-cell ablation of the dual-guidance design. The operating point $\alpha = 4$, $\beta = 2$ was selected on a disjoint 40-item tuning sweep using only independent metrics. Fidelity is BERTScore-F1 and DeBERTa-v3 NLI entailment; engagement-tactic realization an independent LLM judge (0-1); clickbait the independent detector; fluency held-out distilGPT-2 perplexity. Every cell is $n = 300$ items $\times$ 3 tactic configurations.

Table 6: Four-cell dual-guidance ablation on 300 held-out headlines. Clickbait and tactic use independent models; lower perplexity = more fluent.

Cell	tactic (judge)	clickbait (ext.)	BERTScore	NLI	perplexity	paired win vs (0,0)
(0,0) baseline	0.432	0.045	0.282	0.270	211	-
(4,0) engage only	0.472	0.079	0.280	0.219	175	0.610 ($p \sim 1e-13$)
(0,2) brake only	0.399	0.028	0.280	0.324	275	0.444
(4,2) dual	0.474	0.057	0.251	0.213	241	0.597 ($p \sim 3e-9$)

The two guides have the designed, independent, opposite effects: the engagement guide raises tactic realization (+0.04, paired-preferred over the no-guidance baseline in 61% of blind same-item comparisons, $p \sim 1e-13$), while the clickbait brake lowers the independent clickbait score from 0.079 to 0.057 and improves entailment. At the dual operating point the rewrite is blind-preferred over the no-guidance baseline in 59.7% of same-item judgments ($p \sim 3e-9$). Because the prompt already names the target tactic, the (0,0) cell is itself a capable rewriter, so 0.60 is the margin the decoding-time guides add on top of prompting.

4.5 Comparison with decoding-time baselines

We compare FUDGE (alpha=4, beta=2) against a plain prompt-only baseline and the two related decoding-time methods DExperts and GeDi, each with its own Llama-3.2-1B expert/controller and strength sweep, on the same 300 items. Significance is item-matched Wilcoxon signed-rank ($n = 900$).

Table 7: Cross-method comparison at matched independent tactic level. BERTScore/NLI = fidelity; clickbait = independent detector; perplexity = fluency.

Method	tactic (judge)	BERTScore	clickbait (ext.)	perplexity
prompt-only	-	0.366	0.066	141
FUDGE (4,2)	0.500	0.251	0.057	241
DExperts (strong)	0.525	-0.319	0.735	885
GeDi (strong)	0.500	0.055	0.588	1185

At matched independent tactic realization (~ 0.50 ; the three steered methods are statistically indistinguishable on this axis: FUDGE vs DExperts $p = 0.93$, vs GeDi $p = 0.38$), FUDGE is Pareto-dominant on every other axis: BERTScore 0.25 versus DExperts -0.32 and GeDi 0.06 (both $p < 1e-300$), far lower induced clickbait, and 4-5x lower perplexity. DExperts and GeDi match FUDGE's tactic level only by steering so hard that fidelity and fluency collapse (a negative BERTScore means the rewrite barely resembles the source). FUDGE reaches the same targeted control while remaining faithful and fluent, which the single-attribute product-of-experts baselines cannot.

4.6 Human evaluation (LLM-simulated proxy, pending the human study)

The full human protocol (rater packages, blinding, and the kappa / ICC / correlation analysis) is implemented and validated; the human study will be run for the next revision. To exercise the pipeline and provide an interim signal, we fill the rater packages with three independent GPT annotators at raised temperature under strict / lenient / balanced personas. Every number here is an LLM-simulated proxy, not a human result. (C1) Rubric validation over 150 synthetic headlines yields substantial inter-rater agreement (mean Fleiss kappa = 0.75) and a mean intended-vs-consensus tactic F1 of 0.43. (C2) Rewrite quality over 1,500 items across all four methods yields good-to-excellent reliability (ICC(2,k) 0.74-0.93); proxy-rater faithfulness correlates strongly with automatic BERTScore (Spearman rho = 0.78, $n = 1500$), validating the automatic fidelity metric; and the proxy raters independently reproduce the method ranking, rating faithfulness prompt-only (3.78) > FUDGE (3.0-3.3) > GeDi (2.34) > DExperts (2.12) on a 1-5 scale.

4.7 Observed Failure Modes

The qualitative examples in Table 4 are not uniformly successful and several recurring failure modes deserve explicit discussion. First, provocative-question outputs of the form "...: But at what cost to X?" can occasionally slip past the negative guide because their surface lexicon is not exaggerated even though the implied premise is: the interrogative construction shifts the manipulative content into the presupposition, which the prefix-level clickbait scorer is not

explicitly trained to flag. Second, positive guidance targeting referential underspecification can, in rare cases, embellish the source with a novel noun phrase (for example, terms such as "Secrets" or "A New Era") that is not present in the original headline; such outputs increase the attribute-realization score under our own engagement model but reduce semantic fidelity relative to the source. Third, because the clickbait scorer is trained on GPT-4o-mini generations, it may under-flag human-authored clickbait patterns whose surface distribution differs from that of the synthetic corpus; the external-benchmark evaluation of Section 5.1 quantifies this residual gap. We treat these three failure modes as targets for the tactic-specific negative penalties discussed in Section 5 rather than as evidence against the framework as a whole.

5. Conclusion and Future Work

Moreover, clickbait suppression yields headlines that remain semantically faithful to the source while continuing to express the targeted engagement tactic. The quantitative evaluation in Section 4 supports this on 300 held-out headlines: the dual guide is blind-preferred over the no-guidance baseline in 59.7% of same-item judgments ($p \sim 3e-9$), the clickbait brake measurably lowers the independent detector score, and against DExperts and GeDi the method is Pareto-dominant (comparable independent tactic realization at 4-5x lower perplexity and far higher BERTScore fidelity).

The results support the central claim that clickbait is not an inevitable byproduct of engagement optimization. Instead, it emerges when negative constraints are absent. Framing headline generation as a constrained optimization problem provides a practical, interpretable mechanism for navigating the trade-off between informativeness and appeal, with direct relevance to responsible media generation, recommender systems, and editorial AI tools.

5.1 Limitations and Scope of the Guide Models

Second, the neutrality of the Reuters source headlines is quantified rather than merely assumed: the independent detector flags only 0.61% of the source pool (Section 4.3). Third, because both guide models are trained on prefixes from the same GPT-4o-mini pipeline that defines the positive class, their synthetic-test performance overstates generalization; the external evaluation of Section 4.1 measures this directly, showing strong transfer on clean human clickbait (AUROC 0.955) and a moderate but significant drop on the noisier graded corpus (0.740). Three further limitations follow from the new evaluation: (i) the decoding-time guides amplify prompted tactics rather than inject them from scratch, so the framework is evaluated as prompt-plus-guidance; (ii) FUDGE does not realize tactics more strongly than the DExperts/GeDi baselines (the difference is not significant) - its advantage is Pareto, comparable control at higher fidelity and fluency; and (iii) the human evaluation reported here is an LLM-simulated proxy pending the actual human study. All comparisons use item-matched significance with 95% bootstrap confidence intervals.

Several directions naturally extend this study. First, the control framework can be expanded from scalar coefficients to tactic-specific penalties, allowing fine-grained suppression of individual clickbait strategies such as urgency inflation, emotional manipulation, or curiosity gaps. Second, future work should evaluate the approach at scale using human judgments and real-world engagement metrics, enabling calibration of λ_{pos} and λ_{neg} against editorial or platform-specific standards. Third, integrating the control

mechanism into multi-step or reinforcement-based generation loops may further improve robustness under tactic composition. Finally, extending the framework beyond headlines to longer-form content such as summaries or social media posts would test its generality and applicability to broader content moderation and responsible text-generation settings.

Data Availability

The dataset and code are publicly available at: <https://github.com/LinoyHalifa/Clickbait-Research>

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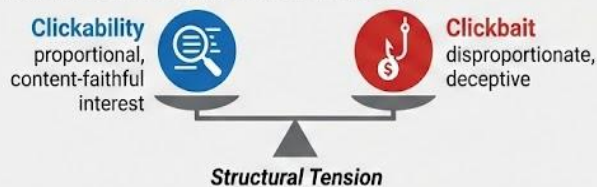
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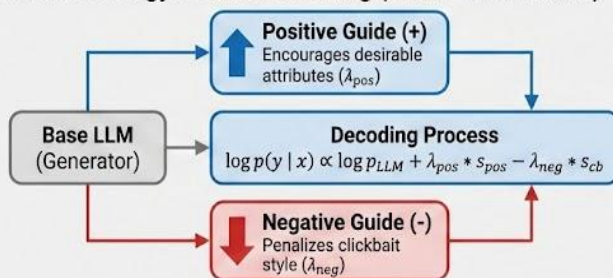
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LLM-Guided Headline Rewriting: Controllable Clickability Enhancement Without Clickbait

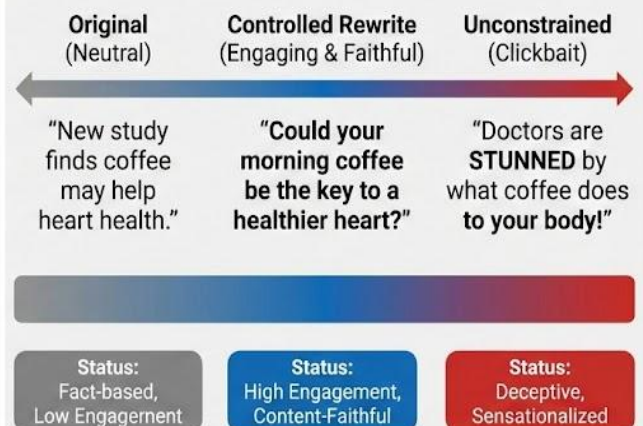
1. The Problem: Clickbait vs. Clickability



2. Methodology: Guided Decoding (FUDGE Framework)



3. Results: Controlled Generation Continuum



Conclusion: Decouples engagement from deception. Clickbait is a **failure of negative constraints**, not an **inevitable** byproduct of optimization. ✨